

Socle Filtration via Littlewood–Richardson Coefficients

Pablo Santos Guerrero

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Abstract

We give a self- contained, rigorous exposition of a Python program that computes the $(k+1)^{\text{st}}$ socle layer of a tensor product of induced modules. The algorithm relies on combinatorial representation theory: partitions, skew shapes, Littlewood–Richardson (LR) coefficients, and Yamanouchi (lattice) words. All mathematical notions are introduced together with the corresponding code fragments; the overall flow of the program is described in detail.

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1 Socle

Table for $\lambda = (1, 1)$, $\mu = (1,)$:

$V^{\emptyset,(1,)} \oplus V^{(1,),(1,)} \oplus V^{(2,),\emptyset} \oplus V^{(2,), (1,)} \oplus V^{(1,1),\emptyset} \oplus V^{(1,1),(1,)}$
$V^{\emptyset,(1,)} \oplus V^{(1,),\emptyset} \oplus V^{(1,),(1,)} \oplus V^{(2,),\emptyset} \oplus V^{(2,), (1,)} \oplus V^{(1,1),\emptyset} \oplus V^{(1,1),(1,)}$

Table 2 for $\lambda = (1, 1)$, $\mu = (1,)$:

$V^{\emptyset,(1,)} \oplus V^{(1,),\emptyset} \oplus V^{(1,),(1,)} \oplus V^{(2,),\emptyset} \oplus V^{(2,), (1,)} \oplus V^{(1,1),\emptyset} \oplus V^{(1,1),(1,)}$
$V^{(1,1),(1,)}$

References

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